

SEMESTER II

Programme	B. Sc. Statistics				
Course Code	STA2CJ101(P)/STA2MN100(P)				
Course Title	Bivariate Data Analysis				
Type of Course	Major				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	HSE level Mathematics/Statistics courses				
Course Summary Objective	To make the student analyze Bi variate data and Examine agreement / strength of variables				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe and interpret bivariate data	U	C	Instructor-created exams / Quiz
CO2	Understand the concept of correlation and interpret their magnitude and direction	U	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Perform simple linear regression analysis to model the relationship between variables	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Identify patterns and trends in bivariate data	U	C	Instructor-created exams / Home Assignments
CO5	Apply effectively in real life situations and do analysis using R software and communicate the results.	Ap	P	One Minute Reflection Writing assignments/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Concept of Bivaraite Data		10	15
	1	Bivariate Data: Definition, Scatter Diagram.		
	2	Contingency tables for discrete data, joint, marginal.		
	3	Curve fitting: Principle of least squares		
	4	fitting of straight line, exponential and power curves using the principle of least squares		
	Sections from References:			
II	Correlation		10	20
	5	Concept, types of Correlation,		
	6	Karl Pearson’s Coefficient of Correlation for grouped and ungrouped data and its properties.		
	7	Spearman’s Rank Correlation		
	8	measures using Discordant and Concordant pairs		
	9	Point Bi serial correlation interpretation of correlation coefficient		
	Sections from References:			
III	Regression		15	20
	10	Concept of Regression		
	11	Distinction between Correlation and Regression		
	12	Linear and Non Linear Regression		
	13	Lines of Regression		
	14	Need of Two lines of Regression		
	15	Regression coefficients		
	16	Properties of Regression Coefficients		
	17	Angle of Regression lines and interpretation		
	Sections from References:			
IV	Partial and Multiple Correlation		10	15
	18	Concepts of Partial and Multiple Correlation Coefficients (three variable cases only).		
	19	Computation of Multiple and Partial Correlation Coefficients		
	20	Properties of Multiple and Partial Correlation Coefficients		
	21	Analysis of Categorical Data: Contingency table,		
	22	Independence & association of attributes.		
	Sections from References:			
V	Open Ended Module: Practical Applications, Case Study and Group Assignments		30	
	1	Practical exercises Hands-on using Software R: Graphical Presentations, Correlation and regression Case study using primary data in the form of Group Assignments and Discussions. Prepare record of at least 10 questions from Module I, II, III and IV using R Package		
	Sections from References:			
Books and References:				
1. Christian Heumann, Michael Schomaker, Shalabh., Introduction to Statistics and Data Analysis,				

Springer Publications,2016

2. S.C.Gupta and V.K.Kapoor., Fundamentals of Applied Statistics, Sultan Chand and Sons

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	3	-	-	-	2	3	2	-	-	-	-
CO 2	-	-	-	-	-	3	3	2	-	-	-	-
CO 3	-	-	-	-	-	2	2	3	-	-	-	-
CO 4	-	-	-	-	-	-	3	3	-	-	1	1
CO 5	-	2	-	3	2	-	2	-	1	-	2	-
CO 6	2	-	2	-	-	3	2	3	-	3	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)